Duration of Immunity Against Pertussis After Natural Infection or Vaccination

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Abstract: Despite decades of high vaccination coverage, pertussis has remained endemic and reemerged as a public health problem in many countries in the past 2 decades. Waning of vaccine-induced immunity has been cited as one of the reasons for the observed epidemiologic trend. A review of the published data on duration of immunity reveals estimates that infection-acquired immunity against pertussis disease wanes after 4–20 years and protective immunity after vaccination wanes after 4–12 years. Further research into the rate of waning of vaccine-acquired immunity will help determine the optimal timing and frequency of booster immunizations and their role in pertussis control.

Key Words: *Bordetella pertussis*, pertussis, immunization, immunity

(*Pediatr Infect Dis J* 2005;24: S58–S61)

Widespread immunization against pertussis was implemented with whole cell (wP) vaccines in the 1940s and 1950s in many industrialized countries. As a result, the incidence of reported pertussis cases decreased 99% in the United States by the 1970s.[1] Despite continuous high immunization coverage rates in the United States and other industrialized countries, pertussis has remained endemic; and the incidence of reported pertussis disease has gradually increased after reaching a nadir in 1976.[2] Waning of vaccine-acquired immunity and decreased opportunities for boosting of immunity because of reduced levels of circulation of *Bordetella pertussis* have been cited as some of the possible reasons for reemergence of pertussis.[3–7]

This article reviews data on the duration of immunity acquired by infection with *B. pertussis*, and duration of immunity acquired by vaccination with wP and acellular pertussis (aP) vaccines.

DURATION OF IMMUNITY ACQUIRED BY INFECTION WITH *B. PERTUSSIS*

Before the implementation of immunization programs, pertussis was widely recognized as a severe disease among children. The majority of reported cases occurred in children, and an estimated 80% of the population suffered from pertussis disease during childhood.[8] Seroprevalence studies in largely unvaccinated populations indicate that almost all children (95%) become infected with pertussis by the age of 19 years.[9] Selected articles describing the duration of potential protection afforded by infection are listed in Table 1.

Adults in the prevaccine era rarely presented with typical forms of pertussis.[13] One study indicated that only 0.26% of a population of 20,405 persons had a second case of pertussis disease.[10] It was therefore postulated that immunity from natural infection was close to lifelong.[11,14] A more recent prospective study of household contacts in a largely unvaccinated population demonstrated that 33% of adult pertussis cases had a history of pertussis during childhood.[11] Computer simulations of the prevaccine era by mathematical modeling estimate that, in 1940, everyone experienced pertussis as a child, 9% of the population had more than 1 typical pertussis episode, and the average adult experienced 2.6 cases of mild pertussis in his/her lifetime.[15] The frequent occurrence of mildly symptomatic *B. pertussis* infections in adults refutes the previously held belief that infection-acquired immunity is lifelong.

Symptomatic reinfections with *B. pertussis* in children have only recently been reported, both in the Netherlands and in Senegal. A prospective cohort study in the Netherlands documented *B. pertussis* infection clinically and by laboratory techniques in both the first and reinfection episode in 4 children.[7] The second episodes of pertussis were milder than

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ISSN: 0891-3668/05/2405-0058

DOI: 10.1097/01.inf.0000160914.59160.41

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TABLE 1. Selected Articles Describing Duration of Protection Acquired by Natural Infection With *Bordetella pertussis*

Author	Year	Participants (n)	Data Source	Estimate of Protection (yr)	Country of Study
Laing and Hay[10]	1902	20,405	Cohort	Near lifelong	U.S.
Gordon and Hood[8]	1951	Not applicable	Review	Near lifelong	Not applicable
Wirsing von König et al[11]	1995	369	Prospective household contact	20	Germany
Miller and Gay[12]	1997	Not applicable	Review/modeling	7–10	U.K.
Versteegh et al[7]	2002	4	Case series	3.5–12	The Netherlands

first infections and may not have been diagnosed outside of a research setting. This study provided well-documented evidence that the duration of infection-acquired immunity in children may be as short as 3.5 years. A study in Senegal of 8419 children documented 2 episodes of confirmed symptomatic pertussis in 137 unvaccinated children (0.02% of 6131 children) and 73 previously vaccinated children (0.03% of 2288 children).[16] The mean time between the first and second infections was 7.1 years [95% confidence interval (95% CI), 6.6–7.6)] in the unvaccinated children and 5.1 years (95% CI 4.5–5.7) in the previously vaccinated children.

The current estimates of the duration of infection-acquired immunity range from 7–10 years[12,16] to 20 years.[11] These widely different results could be the result of differences in levels of circulating *B. pertussis*, surveillance systems and case definitions used.

DURATION OF IMMUNITY ACQUIRED BY wP VACCINATION

Estimates of the duration of immunity acquired after wP vaccination range from 4 to 12 years (Table 2).[5,6,13,14,17–21] These estimates have been derived indirectly from various studies, none of which was designed primarily to study the duration of immunity.

Estimates of duration of immunity after vaccination have been most frequently based on 2 studies.[13,17] Lambert[13] reported on a 1962 outbreak in Michigan, in which 195 cases of *B. pertussis* infection were identified from 474 household members. Vaccination history was collected, and attack rates were calculated, demonstrating that 95% of the attacks had occurred within 12 years since the last dose of wP vaccine. Jenkinson,[17] a private physician in the United Kingdom, studied a semirural community consisting of 11,500 people for a period of 10 years and identified 436 cases of pertussis. Vaccine efficacy was calculated for each age group by excluding the unvaccinated cases and subtracting the attack rate (number of cases divided by the number at risk) from 1. Although children from each age group would have been born in different years, each age group was treated as its own cohort. He estimated that by 4 years after immunization, only 52% of the children still had a clinically protective level of immunity against pertussis.

A recent study based on Australian notification data investigated the effect of age at administration of last vaccine dose on the average age of childhood pertussis cases. Australia introduced a fifth dose of wP vaccine at 4–5 years of age in late 1994. In 1997 the peak rate of disease was among 8- to 9-year-olds, whereas in 2001 the peak rate of disease was in 12- to 13-year-olds. This study thus provides evidence that immunity acquired by wP vaccination wanes 6–9 years after the last dose.[5]

Although vaccine efficacy studies have demonstrated that there is no loss of protective immunity after wP vaccination during the first 2 years after vaccination,[22] asymptomatic infections have been reported to occur within the first year after vaccination with wP vaccine.[23] This indicates that the duration of protective immunity against disease lasts longer than the immunity against infection.

DURATION OF IMMUNITY ACQUIRED BY aP VACCINATION

In the 1990s, aP vaccine efficacy trials were conducted in various countries. In the United States, the first aP vaccine was licensed for use in 1997. Although aP vaccines were only

TABLE 2. Selected Articles Describing Duration of Protection Acquired by wP Vaccine

Author	Year	Participants (n)	Data Source	Estimate of Protection (yr)	Country of Study
Lambert[13]	1965	474	Outbreak	12	U.S. (Michigan)
Jenkinson[17]	1988	436	Clinic population	4	U.K.
CDC[18]	1993	225	Outbreak	4–6	U.S. (Massachusetts)
Ramsay et al[19]	1993	3150	Surveillance data	8	U.K.
Nielsen and Larsen[14]	1994	Unknown	Surveillance data	10	Denmark
He et al[20]	1996	3794	Surveillance data	5–10	Finland and Switzerland
Van Buynder et al[6]	1999	15,286	Surveillance data	5–14	U.K.
Torvaldsen and McIntyre[5]	2003	Unknown	Surveillance data	6–9	Australia

CDC indicates Centers for Disease Control and Prevention.

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recently introduced, some published reports on the duration of immunity are available. Results suggest that the duration of protective immunity after vaccination with aP vaccines is not substantially different from that after vaccination with wP vaccines (Table 3).

A prospective surveillance study of participants of a vaccine efficacy study comparing 2 aP vaccines demonstrated sustained efficacy of 86% for typical pertussis and 76–78% for mild pertussis during the first 6 years of life (ie, >5 years after the last dose).[4] A case-contact study nested in a vaccine efficacy trial demonstrated a difference in vaccine efficacy between the aP and wP vaccines. Whereas in children 18 months–4 years old, the pertussis incidence rate was higher in those vaccinated with aP vaccine than in those vaccinated with wP vaccine [incidence rate ratio (IRR), 1.76; 95% CI 1.33–2.33], there was no difference in the incidence rate in children younger than 18 months old (IRR 1.13; 95% CI 0.66–1.95). This may suggest a longer duration of protective immunity acquired by wP vaccination than by aP vaccination.[24]

In contrast, 2 studies monitoring the long term effectiveness of pertussis vaccines did not find a difference between the duration of immunity after aP (2-component JNIH-6) and wP vaccine (monovalent Wellcome). In a Swedish 10-year follow-up study, 13% of the study population acquired pertussis infection, with a median age of 5.5 years after the last dose, irrespective of type of vaccine.[25] In a German 6-year follow-up study of wP and aP vaccines (produced by Wyeth-Lederle), the calculated efficacy for the 6-year follow-up period was 89% (95% CI 79–94) for the aP vaccine and 92% (95% CI 84–96) for the wP vaccine.[26]

LIMITATIONS IN ADDRESSING DURATION OF IMMUNITY

There are many limitations in our understanding of the duration of immunity after both natural infection and vaccination. No clear serologic marker exists for protective immunity against pertussis. By ~2 years postinfection or postvaccination with either wP or aP vaccine, antibodies have reached barely detectable levels,[27–30] whereas immunity against infection remains.

Studies have not been able to control for levels of circulating *B. pertussis* in the population. This may be important because asymptomatic infections will boost the level of immunity and can thus lead to an overestimation of the duration of protection against symptomatic disease. Pertussis vaccine efficacy studies have demonstrated a decrease in the transmission of *B. pertussis* infection from vaccinees to household contacts.[22,31,32] Observational studies have demonstrated a decrease in *B. pertussis* incidence rates in unvaccinated subgroups when vaccination coverage is >80%.[14,33–35] The indirect effect (or herd immunity) has also not been taken into account in vaccine efficacy studies.

Another limitation of follow-up of vaccine efficacy trials lies in the difficulty of separating vaccine efficacy from waning immunity. If the efficacy of the vaccine is 85%, and there is a 20% age-specific attack rate in a highly vaccinated population, waning of immunity may have contributed to the occurrence of disease in 5% rather than 20% of the population. Nonetheless the potential overall impact on the population remains considerable.

Several factors further contribute to the difficulty in measuring the persistence of immunity and limit comparability between studies: the use of different vaccines (even within the groups of wP or aP vaccines); changes in manufacturing or vaccine contents over time; the presence of different immunization schedules (in timing and/or number of dosages); and utilization of different case definitions, surveillance methods and reporting systems.

CONCLUSION

Despite the limitations in measuring persistence of immunity to pertussis disease after natural infection or vaccination, some common themes have emerged. Protective immunity after infection was probably never lifelong and wanes after 7–20 years. Duration of immunity after either wP or aP immunization does not appear to substantially differ and likely lasts 4–12 years in children. Clear differences

TABLE 3. Selected Articles Describing Duration of Protection Acquired by aP Vaccine

Author	Year	Vaccine Type	Participants (*n*)	Data Source	Duration of Follow-up	Estimate of Protection (yr)	Country of Study
Simondon et al[24]	1997	Pasteur Mérieux Serums and Vaccines (4-component)	4181	Nested case-contact	Up to 4.25 yr	Protection after wP longer than aP	Senegal
Tindberg et al[25]	1999	2-component	207	Follow-up of vaccine efficacy trial	10 yr	5.5	Sweden
Salmaso et al[4]	2001	SmithKlineBeecham and Chiron Biocine (both 3-component)	8432	Vaccine efficacy trial	3 yr	6	Italy
Lugauer et al[26]	2002	4-component	10271	Longitudinal cohort	6 yr	6	Germany

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between immunity after vaccination and disease are difficult to distinguish based on available published data.

The interplay between waning immunity and boosting of pertussis immunity by *B. pertussis* infection and vaccination influences the epidemiology of pertussis and its transmission dynamics in the population. Further research into the rate of waning of vaccine-acquired immunity will help determine the optimal age and frequency of booster immunizations and their role in pertussis control.

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